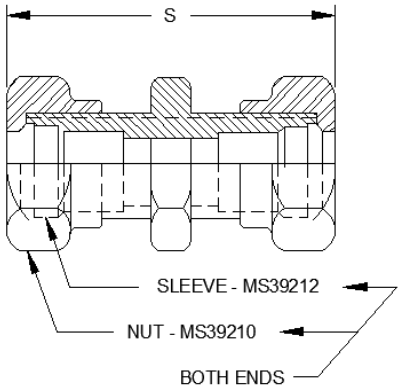
INCH-POUND
MS39201J
1 October 2019
SUPERSEDING
MS39201H
31 January 2018

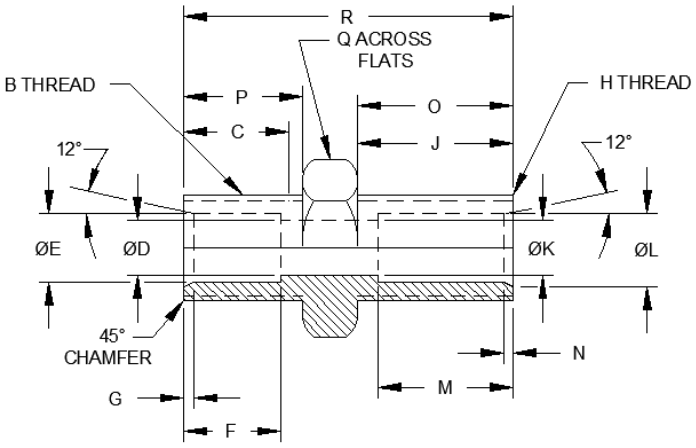
DETAIL SPECIFICATION SHEET

NIPPLE, TUBE-SAFETY SLEEVE COMPRESSION TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-18866.





DETAL VIEW

See notes at end of figure.

FIGURE 1. Nipple, tube compression type.

AMSC N/A

FSC 4730

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Dash Number	Tube OD inch (mm		B Threads	C Thread length inch (mm
-1	.1250 (3.175)	.1250 (3.175)	.3125-24UNF-2A	.219 (5.56)
-2	.1875 (4.763)	.1875 (4.763)	.3750-24UNF-2A	.219 (5.56)
-3	.2500 (6.350)	.2500 (6.350)	.5000-20UNF-2A	.328 (8.33)
-4	.3125 (7.938)	.3125 (7.938)	.5625-18UNF-2A	.328 (8.33)
-5	.3750 (9.525)	.3750 (9.525)	.6250-18UNF-2A	.359 (9.12)
-6	.5000 (12.700)	.4375 (11.113)	.7500-16UNF-2A	.344 (8.74)
-7	.5000 (12.700)	.5000 (12.700)	.7500-16UN-2A	.344 (8.74)
-8	.6250 (15.875)	.6250 (15.875)	.9375-16UN-2A	.359 (9.12)
-9	.7500 (19.050)	.7500 (19.050)	1.0625-16UN-2A	.359 (9.12)
-10	.8750 (22.225)	.8750 (22.225)	1.1875-16UN-2A	.359 (9.12)
-11	1.0000 (25.400)	1.0000 (25.400)	1.3125-16UN-2A	.359 (9.12)
-12	1.2500 (31.750)	1.2500 (31.750)	1.6250-16UN-2A	.478 (12.14)
-13	1.5000 (38.100)	1.5000 (38.100)	1.8750-16UN-2A	.469 (11.91)

Dash Number	D inch (mm)	E ±.005 (0.13)		F inch (mm)	G inch (mm)
		inch	mm		
-1	.094 (2.39)	.135 .139	3.43 3.53	.219 (5.56)	.125 (3.18)
-2	.125 (3.18)	.196 .200	4.98 5.08	.250 (7.54)	.172 (4.37)
-3	.203 (5.16)	.261 .265	6.63 6.73	.297 (7.54)	.203 (5.16)
-4	.234 (5.94)	.324 .328	8.23 8.33	.297 (7.54)	.203 (5.16)
-5	.281 (7.14)	.386 .390	9.80 9.91	.297 (7.54)	.203 (5.16)
-6	.422 (10.72)	.514 .518	13.06 13.16	.313 (7.95)	.203 (5.16)
-7	.422 (10.72)	.514 .518	13.06 13.16	.313 (7.95)	.203 (5.16)
-8	.500 (12.70)	.641 .645	16.28 16.38	.344 (8.74)	.234 (51.94)
-9	.656 (16.66)	.766 .770	19.46 19.56	.344 (8.74)	.203 (5.16)
-10	.719 (18.26)	.891 .895	22.63 22.73	.344 (8.74)	.203 (5.16)
-11	.875 (22.23)	1.016 1.020	25.81 25.91	.406 (10.31)	.203 (5.16)
-12	1.094 (27.79)	1.270 1.274	32.58 32.56	.406 (10.31)	.203 (5.16)
-13	1.344 (34.14)	1.520 1.524	38.61 38.71	.469 (11.91)	.203 (5.16)

See notes at end of figure.

FIGURE 1. Nipple, tube compression type - Continued.

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Dash Number	H Threads	J inch (mm)	K inch (mm)	L ±.005 (0.13)	
				inch	mm
-1	.3125-24UNF-2A	.219 (5.56)	.094 (2.39)	.135 .139	3.43 3.53
-2	.3750-24UNF-2A	.219 (5.56)	.125 (3.18	.196 .200	4.98 5.08
-3	.5000-20UNF-2A	.328 (8.33)	.203 (5.16)	.261 .265	6.63 6.73
-4	.5625-18UNF-2A	.328 (8.33)	.234 (5.94)	.324 .328	8.23 8.33
-5	.6250-18UNF-2A	.359 (9.12)	.281 (7.14)	.386 .390	9.80 9.91
-6	.6875-16NS-2A	.359 (9.12)	.281 (7.14)	.450 .454	13.06 13.16
-7	.7500-16UNF-2A	.344 (8.74)	.422 (10.72)	.514 .518	13.06 13.16
-8	.9375-16UN-2A	.359 (9.12)	.500 (12.70)	.641 .645	16.28 16.38
-9	1.0625-16UN-2A	.359 (9.12)	.656 (16.66)	.766 .770	19.46 19.56
-10	1.1875-16UN-2A	.359 (9.12)	.719 (18.26)	.891 .895	22.63 22.73
-11	1.9375-16UN-2A	.359 (9.12)	.875 (22.23)	1.016 1.020	25.81 25.91
-12	1.6250-16UN-2A	.438 (11.13)	1.094 (27.79)	1.270 1.274	32.58 32.56
-13	1.8750-16UN-2A	.469 (11.91)	1.344 (34.14)	1.520 1.524	38.61 38.71

Dash Number	M inch (mm	N inch (mm	O inch (mm	P inch (mm
-1	.219 (5.56)	.125 (3.18)	.281 (7.14)	.281 (7.14)
-2	.250 (6.35)	.172 (4.69)	.281 (7.14)	.281 (7.14)
-3	.297 (7.54)	.203 (5.16)	.406 (10.31)	.406 (10.31)
-4	.297 (7.54)	.203 (5.16)	.406 (10.31)	.406 (10.31)
-5	.297 (7.54)	.203 (5.16)	.438 (11.13	.438 (11.13)
-6	.297 (7.54)	.203 (5.16)	.438 (11.13	.438 (11.13
-7	.313 (7.95)	.203 (5.16)	.438 (11.13	.438 (11.13
-8	.344 (8.74)	.234 (5.94)	.453 (11.51)	.453 (11.51)
-9	.344 (8.74)	.203 (5.16)	.453 (11.51)	.453 (11.51)
-10	.344 (8.74)	.203 (5.16)	.453 (11.51)	.453 (11.51)
-11	.406 (10.31)	.203 (5.16)	.453 (11.51)	.453 (11.51)
-12	.406 (10.31)	.203 (5.16)	.547 (13.98)	.547 (13.89)
-13	.469 (11.91)	.203 (5.16)	.563 (14.30)	.563 (14.30)

See notes at end of figure.

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Dash Number	Q inch (mm	R inch (mm	S inch (mm
-1	.438 (11.13)	.750 (19.05)	1.313 (33.35)
-2	.500 (12.70)	.766 (19.46)	1.313 (33.35)
-3	.625 (15.88)	1.000 (25.40)	1.750 (44.45)
-4	.688 (17.48)	1.000 (25.40)	1.750 (44.45)
-5	.750 (19.05)	1.094 (27.79)	1.938 (49.23)
-6	.875 (22.23)	1.094 (27.79)	1.969 (50.01)
-7	.875 (22.23)	1.094 (27.79)	2.000 (50.80)
-8	1.125 (28.58)	1.250 (31.75)	2.250 (57.15)
-9	1.250 (31.75)	1.250 (31.75)	2.375 (60.33)
-10	1.375 (34.93)	1.313 (33.35)	2.438 (61.93)
-11	1.500 (38.10)	1.313 33.35)	2.438 (61.93)
-12	2.000 (50.80)	1.500 (38.10)	2.875 (73.03)
-13	2.250 (57.15)	1.500 (38.10)	2.938 (74.63)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Tolerances: Unless otherwise specified, decimals ±.016 (0.41mm), degrees ±2°.
4. Threads shall be in accordance with FED-STD-H28.
5. Break all sharp edges; remove all burrs and slivers.
6. This illustration is for identification and is not intended to restrict designs or shapes not dimensioned.

FIGURE 1. Nipple, tube compression type - Continued.

REQUIREMENTS:

Fittings shall be as specified on figure 1 and table I.

Unless otherwise specified materials shall be in accordance with MIL-DTL-18866 and table I.

The part or identifying number (PIN) is for the complete item with MS39210 nuts and MS39212 sleeves.

Maximum operating pressure. See MIL-DTL-18866.

Finish. Finishes shall be as specified in table I. All plating's shall be capable of meeting a minimum of 96 hours salt spray test in accordance with ASTM B117. The fittings shall show no evidence of red corrosion after 96 hours of salt spray. Fluid passages, other openings and internal threads shall not be subject to the plating thickness requirement and may have bare areas provided they are protected with a light film of oil.

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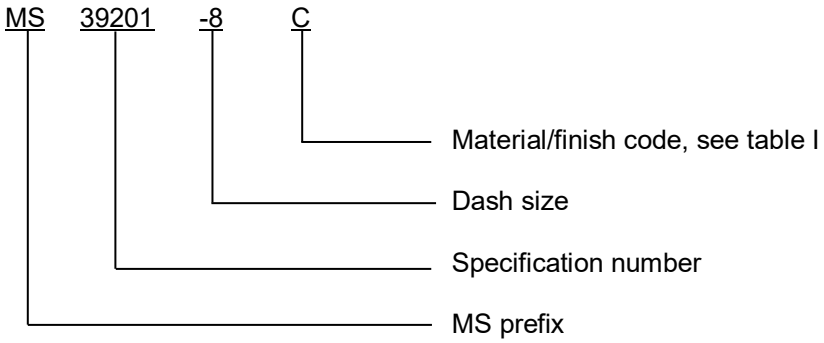
Table I. Material and finish designators.

Material finish designator	Material	Finish
C	Steel bar, type 1117, 1118 or 1137 in accordance with ASTM A29/A29M	Carbon steel: Cadmium in accordance with SAE-AMS-QQ-P-416, type II, class 3, 200µ inches to 300µ inches (5.08 µm to 7.62 µm) thick. 1/ 5/ 7/
CZ	Steel bar, type 1117, 1118 or 1137 in accordance with ASTM A29/A29M	Zinc plating in accordance with ASTM B633; type III or V, Fe/Zn 8. 4/ 5/
F	Steel	Cadmium in accordance with SAE-AMS-QQ-P-416, type II, class 3, 200µ inches to 300µ inches (5.08 µm to 7.62 µm) thick. 1/ 5/ 7/
H	Steel	Zinc-Aluminum in accordance with ASTM F1136/F1136M, grade 3, NC.
J	Steel	Zinc-nickel in accordance with SAE-AMS2417, type 2, grade B. 2/ 3/
M	Nickel-copper alloy UNS N04400	No additional finish.
N	High-chromium nickel alloy UNS N06690	No additional finish.
P	Steel	Zinc phosphate finish in accordance MIL-DTL-16232 type Z, class 1. 4/
R	Steel	Zinc plating in accordance with ASTM B633; type III or V, Fe/Zn 8. 4/ 5/
S	Corrosion resistant steel	Corrosion resistant steel: Passivate in accordance with SAE AMS2700, type 6 or 7, free machining alloy UNS S30300 method 1, type 2 or 4.
T	Titanium	Anodize in accordance with SAE-AMS2488 type 2. 6/ 8/
TF	Titanium	Fluoride phosphate in accordance with SAE-AMS2486. 6/ 8/
Z	Steel	Zinc plating in accordance with ASTM B695, type II, class 8 4/ 5/
ZC	Steel	Zinc may be any zinc plating from PIN codes CZ, J, and Z with a colored chromate coating. 4/ 5/

1/ Embrittlement test need not be run.
2/ Hexavalent chromium free. Finish shall be RoHS compliant.
3/ The zinc-nickel alloy plate shall contain 12% to 16% nickel. The coating thickness shall be 315µ inches (8µm) minimum.
4/ Not for use in aircraft. Requires approval from the Program Office for all applications. Zinc-plated materials should be avoided in hydraulic systems with water-glycol fluids. Reference ASTM MNL37 - Fuels and Lubricants Handbook.
5/ These finishes have a chromate conversion coating which includes hexavalent chromium and are not recommended for Army ground systems.
6/ A pretreatment, a modification of the fluoride treatment, or a post treatment shall be applied so the final color of the fittings shall be similar to SAE-AMS-STD-595 colors 36076 through 36293.
7/ Cadmium shall not be used in oxygen or potable water systems. Cadmium is prohibited from use in NAVSEA-owned shipboard systems in accordance with the requirements of T9070-AL-DPC-020/077-2, unless otherwise approved by NAVSEA.
8/ Titanium shall not be used in oxygen or potable water systems.

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PIN example: The PIN consists of the MS prefix, specification number, dash number for size code, and a material finish design.



MS39201-8C indicates a 5/8 nipple, steel bar with cadmium plating.

Cadmium is not recommended. To the users of this document, it is recommended that the use of carbon steel material with cadmium plating be used only when other materials and finishes specified in this document cannot meet performance requirements.

Cadmium is prohibited from use in NAVSEA-owned shipboard systems in accordance with the requirements of T9070-AL-DPC-020/077-2, unless otherwise approved by NAVSEA.

Class I and II ozone depleting substances (ODS) shall not be used in MS39201 or any referenced procedures.

Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Referenced documents shall be of the issue in effect on date of invitations for bid.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extent of the changes.

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Referenced documents. In addition to MIL-DTL-18866, this document references the following:

FED-STD-H28	SAE-AMS-QQ-P-416	SAE-AMS-STD-595/36231
MIL-DTL-16232	SAE-AMS-STD-595/36076	SAE-AMS-STD-595/36251
MS39210	SAE-AMS-STD-595/36081	SAE-AMS-STD-595/36270
MS39212	SAE-AMS-STD-595/36099	SAE-AMS-STD-595/36280
ASTM MNL37	SAE-AMS-STD-595/36118	SAE-AMS-STD-595/36293
ASTM A29/A29M	SAE-AMS-STD-595/36134	SAE-AMS2417
ASTM B117	SAE-AMS-STD-595/36152	SAE-AMS2486
ASTM B633	SAE-AMS-STD-595/36170	SAE-AMS2488
ASTM B695	SAE-AMS-STD-595/36173	SAE-AMS2700
ASTM F1136/F1136M	SAE-AMS-STD-595/36176	T9070-AL-DPC-020/077-2

CONCLUDING MATERIAL

Custodians:
Army - AT
Navy - AS
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2019-075)

Review activities:
Army - AR
Navy - MC, SA

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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